An obstruction theoretic approach to integrated information

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Abstract

We define the integrated cohomology groups of physical systems, and show that integrated information can be detected by the non-vanishing of H^1 obstruction classes that measures the failure of the cause-effect properties of the system to decompose into smaller parts. This gives a completely functorial notion of integrated information, as suggested by Tull–Kleiner, and allows us to further define higher order integration in a very natural way.

1 Introduction

Integrated Information Theory (IIT) is a mathematical description of the existing consciousness of a system, developed mainly by Tononi and his collaborators, CITE . The theory is centered around the computation of Φ, which is a measure of the *integrated information* in the system. Informally, a system S in a state s, has integrated information, captured by $\Phi(S, s) > 0$, if the information generated by the entire system's cause-effect structure in state s is greater than the information generated by the combined cause-effect structure of partitions of the system. The formal mathematical framework for computing Φ is rigorous and technical, and uses detailed features of probability theory and measure theory, see CITE

In the present paper we wish to study integrated information through the lens of obstruction theory, which is a mathematical framework studying certain obstruction classes that measures whether a property can be extended through defined levels of sub-structure. They live in mathematical objects called *cohomology groups*, which are abstract formulations of the amount of holes a system has in different dimensions. Obstruction theory was originally developed by Stiefel and Whitney to prove the non-existence of certain vector-fields on manifolds CITE , but has later been applied in a plethora of mathematical contexts CITE .

The conceptual similarity between these two ideas is also extendable to a formal description, which is the central aim of this short paper. Building on work by Kleiner and Tull on a categorical description of integrated information theory, see CITE , we assign a simplicial complex $N^\bullet$ to an IIT system S, and use this to build cohomology groups valued in a sheaf $\mathcal{F}$ that incorporates the cause-effect dynamics of S. Intuitively, we wish to build a mathematical structure whose n-dimensional holes represent n-th order integration, or causal connectedness. We then define a natural obstruction class Δ^ψ for any attempted partition ψ of S into subsystems, which we show implies properties for IIT. In particular, if there is a non-trivial obstruction class, then the system has integrated information. These classes will detect whether the global cause-effect dynamics of S can be "glued together" from the local dynamics of its decompositions. This gluing is described by the presheaf, and the failure to glue is measured by the cohomology.

Theorem 1.1. *Let s be a state of a system S. If there is a non-zero obstruction class $\Delta^\Psi \in H^1(N^\bullet; \mathcal{F})$, then $\Phi(S, s) > 0$.*

The upshot of this work is that one can study integrated information from a new lens, and apply fundamental ideas from algebraic topology and homological algebra to study phenomena in consciousness. In particular, our theory gives a much more general approach to integrated information, that side steps a lot of technicalities. We expect that our methods can be combined with topological data analysis, see CITE for an introduction, to locate minimum partitions (MIP) and connect integrated information to new developments in the relationship between topology and information theory, see for example CITE and CITE .

In the last section we consider two natural extensions of our framework: the size of Φ and higher order in-

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tegration. In particular, we construct a cohomological graded integrated information value Φ^k_{coh}, which we conjecture is related to classical integrated information Φ. Our above theorem proves that

$$\Phi = 0 \implies \Phi^1_{coh} = 0,$$

but we suspect that the converse is not true due to the possible non-vanishing of the higher cohomology classes.

We make no claim surrounding the validity of IIT to explain consciousness and the properties of experience, and make no contributions to the ongoing debates CITE . We only wish to connect some of IIT's mathematical properties to other mathematical concepts known to measure similar features. We view this work not as a new standard for IIT, but as a starting point to explore some of the rich mathematical structures related to it.

2 IIT systems

In CITE , Tull and Kleiner develop a category theoretic foundation for integrated information theory, based on symmetric monoidal categories with some extra structure that allows for the modeling of cause-effect structures. We will use this as a backdrop for our cohomology theory, hence we quickly review their setup.

2.1 Process theories and decompositions

Definition 2.1. A *process theory* is a symmetric monoidal dagger category $(\mathcal{C}, \otimes, \mathbb{1}, \dagger)$, usually denoted simply by $\mathcal{C}$, together with two maps $q_S \colon S \longrightarrow \mathbb{1}$ and $w_S \colon \mathbb{1} \longrightarrow S$ for any $S \in \mathcal{C}$ satisfying monoidality and standard naturality conditions. Objects in $\mathcal{C}$ are interpreted as system types, and morphisms as physical processes. The dagger $\dagger \colon \mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{C}$ is interpreted as a time-reversal operation.

Morphisms of the form $\mathbb{1} \longrightarrow S$ will be reffered to as *states*, and we will write $\mathrm{St}(S)$ for the collection of states of S. Similarly, morphisms $S \longrightarrow \mathbb{1}$ will be referred to as *effects* and morphisms $\mathbb{1} \longrightarrow \mathbb{1}$ *scalars*. The designated maps q and w are called the *discarding effect* and the *noise state* respectively, and are interpreted as such.

Definition 2.2. A morphism $f \colon S \longrightarrow S'$ is *causal* if it preserves the discarding effect,

$$q_{S'} \circ f = q_S,$$

and similarly *cocausal* if it preserves the noise state.

Remark 2.3. Tull-Kleiner does not assume $\mathcal{C}$ to be a dagger category for their most general setup, but for concretely tying the categorical version to the IIT algorithm of CITE , as we do in Section 3.3, this is required.

Example 2.4. The standard example of a process theory is the category of classical probabilistic processes. This is the category Class of finite sets, where a morphism $S \longrightarrow S'$ assigns to an element $s \in S$ an 'unnormalized probability distribution' of elements in S', which are maps $S \times S' \longrightarrow \mathbb{R}^+$. The time reversal operation is given by $f^\dagger(s, s') = f(s', s)$. Composition of two maps $f \colon S \times S' \longrightarrow \mathbb{R}^+$ and $g \colon S' \times S'' \longrightarrow \mathbb{R}^+$, is defined by

$$\begin{aligned} f \circ g \colon S \times S'' &\longrightarrow \mathbb{R}^+ \\ (s, s'') &\longmapsto \sum_{s' \in S'} f(s, s') g(s', s''). \end{aligned}$$

The monoidal structure is the cartesian product, meaning the unit $\mathbb{1}$ is the singleton set. Product of morphisms is given by

$$f \otimes g(s, s'')(s', s''') = f(s, s') g(s'', s''').$$

The discarding process q_S is the unique process with $q_S(s) = 1$ for all $s \in S$, meaning a process is causal whenever it is stochastic. The noise state is the uniform probability distribution with $w_S(s) = \frac{1}{|S|}$.

Definition 2.5. Let $\mathcal{C}$ be a process theory, and S an object in $\mathcal{C}$. A *decomposition* of S is a pair of objects $A, B \in \mathcal{C}$ together with a causal isomorphism $\psi \colon S \simeq A \otimes B$. We denote decompositions by (A, B, ψ), or sometimes simply by (A, B).

Any object $S \in \mathcal{C}$ has a decomposition $(S, \mathbb{1})$, which we call the *trivial* decomposition. We will further say that two decompositions (A, B, ψ) and (A', B', ψ') are *equivalent* if there are causal isomorphisms $f \colon A \simeq A'$ and $g \colon B \simeq B'$, such that

$$\psi' = (f \otimes g) \circ \psi.$$

The *complement* decomposition of (A, B) is the same decomposition, just in the opposite order; it is denoted $(A, B)^\perp = (B, A)$. Note that, if two decompositions are equivalent, then so are their complements.

Definition 2.6. Let $\mathbb{D}(S)$ denote the set of equivalence classes of decompositions of S. The operation $(-)^{\perp}$ acts on $\mathbb{D}(S)$, and we define a *decomposition set* of S to be a subset $\mathbb{D} \subseteq \mathbb{D}(S)$ that contains the trivial decomposition and is closed under $(-)^{\perp}$.

Definition 2.7. Given a decomposition set $\mathbb{D}$ of S, and a decomposition $(A, B, \psi) \in \mathbb{D}$, we define the *restricted* decomposition set $\mathbb{D}_{|A}$ to be the set of decompositions of A that can be extended to a decomposition of S in $\mathbb{D}$. Furthermore, we define the *restriction* of a state $s\colon \mathbb{1} \longrightarrow S$ to A to be the state defined by

$$s_{|A} = (\mathrm{Id}_A \otimes q_B) \circ \psi \circ s\colon \mathbb{1} \longrightarrow A,$$

and similarly for B.

Intuitively, the restriction is defined by identifying S with its decomposition, and then discarding the factor we don't want to consider. By functoriality and naturality, such restrictions depend only on the equivalence class in $\mathbb{D}(S)$, and not on the particular representative.

2.2 System types and directional cuts

To define generalized integrated information theories, we need a notion of systems to study. These are objects in the ambient category $\mathcal{C}$ together with the extra structure defined in the previous section.

Definition 2.8. A *system* in $\mathcal{C}$ is a triple $(S, \mathbb{D}, T)$, where $S \in \mathcal{C}$, $\mathbb{D}$ is a decomposition set of S, and $T\colon S \longrightarrow S$ is a causal process, interpreted as a *time evolution*. It is assumed that also $T^{\dagger}$ is causal.

Example 2.9. The most basic example is the *trivial* system $(\mathbb{1}, \{\mathbb{1} \otimes \mathbb{1}\}, \mathrm{Id}_{\mathbb{1}})$.

A state of a system $(S, \mathbb{D}, T)$ is a state of S. Given such a state s and a decomposition $(A, B) \in \mathbb{D}$, CITE constructs a *subsystem* $(A, \mathbb{D}_{|A}, T_{|A})$, where the restricted time evolution is defined by the composition

$$A \xrightarrow{1 \otimes s_{|B}} A \otimes B \xrightarrow{T} A \otimes B \xrightarrow{1 \otimes q_A} A.$$

These subsystems are again independent of the class representative in $\mathbb{D}$.

Given a system $(S, \mathbb{D}, T)$ and a decomposition $(A, B, \psi) \in \mathbb{D}$, one needs to be able to form systems that disconnects certain causal connections in the total system. IIT 3.0 CITE does this by introducing *cut systems*, that directionally disconnects causal flow from the two parts. IIT 4.0 CITE does this in a slightly different manner, by replacing parts of the inputs of the time evolutions with independent noise. We have included versions of IIT 4.0's disconnections in arbitrary system states for reference and intuition, but we remark that the actual implementation of such disconnections does not matter much, as long as they are functorial.

Definition 2.10. Given a decomposition $(A, B, \psi) \in \mathbb{D}$, the *left cut* replaces the contribution of B to the time evolution T with the noise state, meaning that only the contribution from A matters. Formally, it is the system $(S, \mathbb{D}, T^{\psi,\leftarrow})$, where the new time evolution operator is defined as

$$T^{\psi,\leftarrow}\colon A \otimes B \xrightarrow{1 \otimes w_B q_B} A \otimes B \xrightarrow{T} A \otimes B.$$

In spirit this forces the time evolution to retain all self-influences from A, and influences from A to B, but removes all influences from B to A. The arrow in the notation is meant to denote the direction that is removed.

We define the *right cut* similarly by replacing T with

$$T^{\psi,\rightarrow}\colon A \otimes B \xrightarrow{w_A q_A \otimes 1} A \otimes B \xrightarrow{T} A \otimes B,$$

and the *double cut* by the diagram

$$\begin{tikzcd} A \otimes B \arrow[rr, "T^{\psi,\leftrightarrow}"] \arrow[d, "1 \otimes w_A \otimes w_B \otimes 1"'] & & A \otimes B \\ A \otimes B \otimes A \otimes B \arrow[rr, "T \otimes T"] & & A \otimes B \otimes A \otimes B \arrow[u, "1 \otimes q_B \otimes q_A \otimes 1"'] \end{tikzcd}$$

Remark 2.11. The double cut corresponds to Tull–Kleiner's bidirectional cut, where one evolves the system as if A and B were completely disconnected from each other.

We will use the notation $T^{\psi,\alpha}$, with $\alpha \in \{\leftarrow, \rightarrow, \leftrightarrow\}$ to denote a general disconnection based on a decomposition $\psi\colon S \simeq A \otimes B$. As we did above with the restricted time evolution, we also get definitions of restricted cuts due to the functoriality of discarding and noise; these are denoted $T^{\psi,\alpha}_{|A}$.

3 Integrated cohomology and obstruction classes

In this section we present the more novel parts of this paper, which is a cohomology theory for IIT systems. We then build an obstruction theory for decomposing a physical system based on this cohomology theory, and show that it detects the existence of integrated information. Lastly we extend this to naturally define higher order integration, often referred to as higher *synergy*.

3.1 Prerequisites

In essence, cohomology groups are mathematical constructions that measure the amount of holes of different dimensions a space or other mathematical object has. It is generally defined using homological algebra, see CITE for an introduction.

Definition 3.1. A *cochain complex* $C^\bullet$ is a sequence of abelian groups

$$\cdots \longrightarrow C^{i-1} \xrightarrow{\delta^{i-1}} C^i \xrightarrow{\delta^i} C^{i+1} \longrightarrow \cdots$$

such that $\delta^{i-1} \circ \delta^i = 0$ for all i.

The the elements of $C^\bullet$ are called the i-cochains. An i-cochain in the image of the map δ^{i-1} is called a coboundary, while an i-cochain in the kernel of δ^i is called a cocycle. The equation $\delta^{i-1} \circ \delta^i = 0$ implies that $\mathrm{Im}(\delta^{i-1}) \subseteq \mathrm{Ker}(\delta^i)$, which is the fundamental fact used to define cohomology – in words, any coboundary is also a cocycle.

Definition 3.2. Given a cochain complex $C^\bullet$, its i'th *cohomology group* is defined as the quotient group

$$H^i(C^\bullet) := \mathrm{Ker}(\delta^i)/\,\mathrm{Im}(\delta^{i-1}).$$

Elements of $H^i(C^\bullet)$ are called *cohomology classes.*

One of the central goals of algebraic topology and algebraic geometry is to functorially assign cochain complexes to topological and geometric objects and compute the resulting homology groups. For a category of interest $\mathcal{C}$, for example topological spaces, manifolds or schemes, such assignments defines functors into the category of graded abelian groups $H^*\colon \mathcal{C} \longrightarrow \mathrm{Ab}_*$ called *cohomology theories.* Examples include *simplicial cohomology* CITE , *de Rham cohomology* CITE and *coherent cohomology* CITE .

3.2 Integrated cohomology

Given a system $(S, \mathbb{D}, T)$ and a global state $s\colon \mathbb{1} \longrightarrow S$, we will denote by $\mathrm{Sub}_s(S)$ the category of subsystems of S, defined as the poset category associated to the poset structure in CITE . In other words, $\mathrm{Sub}_s(S)$ is the category consisting of subsystems as objects, and morphisms being single comparison maps by the preorder structure.

The overall idea of our approach can be described as follows:

1. first, define a presheaf $\mathcal{F}$ on $\mathrm{Sub}_s(S)$ that describes the cause-effect dynamics of subsystems of S;
2. second, define the integrated cohomology of S to be the presheaf-cohomology of the nerve of $\mathrm{Sub}_s(S)$ valued in $\mathcal{F}$;
3. third, define natural obstruction classes in $H^1(N^\bullet(\mathrm{Sub}_s(S)); \mathcal{F})$;
4. and lastly, prove that these obstruction classes detect the existence of integrated information in S.

There are several possible choices for what presheaf $\mathcal{F}$ to use, depending on how categorical or how system specific one wants to be. If one focuses on actual IIT 4.0 constructions, then perhaps a probability based approach using intrinsic information and intrinsic difference would be a good strategy. We have, however, opted for a more categorical and general approach. This is in part to give generalized integrated information functorial properties, as discussed by Tull–Kleiner in CITE .

As we want to have access to the time evolutions T and $T^{\psi,\alpha}$ for subsystems, it is natural to consider the *endomorphisms* as a natural starting position for the presheaf. As we need to be able to understand the *difference* between T and $T^{\psi,\alpha}$, we also need access to a subtraction operation. The standard way to do this is as follows.

Definition 3.3. The *dynamic presheaf* on $\mathrm{Sub}_s(S)$ is defined on objects as the free abelian group on the associated endomorphisms,

$$\begin{aligned}\mathcal{F}\colon\ \mathrm{Sub}_s(S)^{\mathrm{op}} &\longrightarrow \mathrm{Ab}\\ A &\longmapsto \mathbb{Z}[\mathrm{End}(A)]\end{aligned}$$

Given a subsystem $U \leqslant A$, we define for any endomorphism $f\colon A \longrightarrow A$ a restriction map

$$\rho_U^A(f) = (\mathrm{Id}_U \otimes q_V) \circ f \circ (\mathrm{Id}_U \otimes w_V),$$

where V is the complement of U, i.e. the object such that $A \simeq U \otimes V$, which exists by the definition of a subsystem. We then linearly extended this to $\mathbb{Z}[\mathrm{End}(A)]$, making $\mathcal{F}$ a presheaf.

Remark 3.4. Our setup is formally analogous to Abramsky–Brandenburger's sheaf theoretic description of contextuality, see CITE and CITE , but differs in the added dynamics of the systems. However, we feel that the similarities are enticing, and that the possible interactions are worth further investigation.

Remark 3.5. One could also naturally choose to use the actual state $s_{|V}$ instead of the noise state w_V in the definition of the restriction maps. Doing this would mean that the obstruction theory detects state-dependent integration, and not the intrinsic integration of the system. We believe that studying what happens upon making this change is an interesting avenue of future research.

Definition 3.6 (CITE)**.** Let $\mathcal{C}$ be a category. The *nerve* of $\mathcal{C}$ is the simplicial complex $N^\bullet(\mathcal{C})$ defined by having a 0-simplex for each object in $\mathcal{C}$, a 1-simplex for each morphism in $\mathcal{C}$, a 2-simplex for every pair of composable morphisms, and generally an k-simplex $c_0 \to c_1 \to \cdots \to c_k$ for any k-tuple of composable morphisms. For $0 < i < k$ the i'th face maps f_i are given by composing the two maps, while for $i = 0, k$ it is given by omitting the first or last object respectively. The i'th degeneracy map d_i is given by inserting an identity map at the i'th object.

The nerve of a category is a way of encoding how all objects in the category are related to each other via a combinatorial space. Given a presheaf $\mathcal{F}$ on the category $\mathcal{C}$, one gets a cochain complex

$$C^i(\mathcal{C};\mathcal{F}) = \prod_{c_0 \to \cdots \to c_i} \mathcal{F}(c_0),$$

where the coboundary map $\delta^k\colon C^k(\mathcal{C};\mathcal{F}) \longrightarrow C^{k+1}(\mathcal{C};\mathcal{F})$ is defined on an k-cochain θ by the alternating sum

$$\delta^k(\theta) = \sum_{i=0}^{k+1} (-1)^i d_i(\theta).$$

More explicitly, on a $k+1$-simplex $\sigma = c_0 \to \cdots \to c_k$ we have

$$d_0(\theta)(c_0 \to \cdots \to c_k) = \rho_{c_0}^{c_1}(\theta)(c_1 \to \cdots \to c_k)$$

and similarly for k. For $i > 0$ we have

$$d_i(\theta)(c_0 \to \cdots \widehat{c_i} \to \cdots \to c_k),$$

where $\widehat{c_i}$ denotes the i'th object being omitted.

Remark 3.7. The above definition of the coboundary operator is completely analogous to the boundary operator for Čech cohomology, see CITE .

Important for us will be the coboundary of a 0-cochain θ, which is given on a 1-simplex $\sigma\colon U \longrightarrow V$ as

$$\delta^0\theta(\sigma) = \rho_U^V(\theta(V)) - \theta(U)$$

We can now define the cohomology theory that we will use to define the obstruction classes.

Definition 3.8. Given a system $(S, \mathbb{D}, T)$ and a state $s\colon \mathbb{1} \longrightarrow S$, we define the *integrated cohomology* of $(S, \mathbb{D}, T)$ to be the cohomology of the nerve of $\mathrm{Sub}_s(S)$ evaluated in the dynamic presheaf $\mathcal{F}$. In other words, the i'th integrated cohomology group is defined by

$$H^i_{int}(S) = H^i(N^\bullet(\mathrm{Sub}_s(S)); \mathcal{F}) = \mathrm{Ker}(\delta^i)/\,\mathrm{Im}(\delta^{i-1}).$$

Remark 3.9. It is important to note that this is not simply the topological nerve cohomology of $\mathrm{Sub}_s(S)$, as this will almost always be trivial, see for example CITE . The important part here is the use of the presheaf $\mathcal{F}$, which makes this a highly non-trivial cohomology theory.

The intuition is as follows: The nerve of $\mathrm{Sub}_s(S)$ gives us a geometric construction of *where* integrated information can be tested. It consists formally of chains of substates of S of all lengths, with topological structure that understands how these chains are related and connected. The presheaf gives us *what* is measured on every level – in our case the system's causal dynamics. The cohomology gives us information about which local dynamics that can be glued together to form a global dynamic. If gluing fails, then there is some level of the system where its global dynamics cannot be described in terms of local dynamics, which is the essence of the system having integrated information. In particular, a 1-cocycle measures the failure of a pairwise compatability, which we use to define obstruction classes.

Definition 3.10. Given a decomposition ψ and a direction $\alpha \in \{\to, \leftarrow, \leftrightarrow\}$, its associated *obstruction class* is defined as the cohomology class of the 1-cochain $\Delta^{\psi,\alpha}$ defined on 1-simplicies $\sigma \colon U \longrightarrow V$ as

$$\Delta^{\psi,\alpha}(\sigma) = \rho_U^V(T_{|V}) - T_{|U}^{\psi,\alpha}.$$

The central claim of this paper is that these classes detect whether a system S has integrated information or not. Hence, checking that these classes vanish proves that integrated information is non-zero, which might be much easier than computing Φ, which can be a very hard task – see CITE .

Lemma 3.11. *The obstruction classes $\Delta^{\psi,\alpha}$ are cocycles.*

Proof. By definition we need to prove that $\delta^1\Delta^{\psi,\alpha} = 0$. Let $\sigma = U \to V \to W$ be the 2-simplex consisting of 1-simplicies $\sigma_1 \colon U \to V$ and $\sigma_2 \colon V \to W$. We have

$$\delta^1\Delta^{\psi,\alpha}(\sigma) = \rho_V^W(\Delta^{\psi,\alpha}(\sigma_2)) - \Delta^{\psi,\alpha}(\sigma_2 \circ \sigma_1) + \Delta^{\psi,\alpha}(\sigma_1).$$

□

Remark 3.12. The reader paying minute attention has perhaps noticed that mechanisms and purviews have yet to be featured in this framework. Our approach to measuring IIT is structurally somewhat different, as instead of starting with mechanisms and purviews and then aggregate local information upwards, we have the global structure as a reference and let mechanisms and purviews be derived observables. Hence they do not arise as primitive data, but as local test probes for evaluating the obstruction classes. We will, however, need to define and use these to relate our obstruction classes to the integrated information of S, as we will do in the following section.

3.3 Obstruction classes and generalized IIT

The goal of this section is to prove the following theorem.

Theorem 3.13. *Let s be a state of a system $(S, \mathbb{D}, T)$. If there is no integrated information in S, then all of the decomposition obstruction classes vanish. In other words,*

$$\Phi(S, s) = 0 \implies [\Delta^{\psi,\alpha}] = 0$$

for all A, ψ and α. In particular, the existence of non-vanishing obstruction classes implies $\Phi(S, s) > 0$, and hence that the classes $[\Delta^{\psi,\alpha}]$ form obstructions to the decomposition of causal structures.

In order to do so we need to be able to compute and understand Φ. We will use the method of *generalized IIT's* as developed by Tull–Kleiner in CITE , also formalized in the categorical setting in CITE . This ensures that our theory works not only for classical IIT, but also quantum IIT and other variants that one would like to study, CITE .

Definition 3.14. A *mechanism-purview pair* is a pair of subsystems of a given system $(S, \mathbb{D}, T)$, depending on the same global state $s \colon \mathbb{1} \longrightarrow S$.

Intuitively a mechanism describes the subset of S that we are "letting make a difference", and the purview describes the subset of S that is affected.

Definition 3.15. Given a mechanism $(M, \mathbb{D}_{|M}, T_{|M})$ and a purview $(P, \mathbb{D}_{|P}, T_{|P})$, we define the *cause* to be process

$$\mathrm{cau} \colon M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P,$$

where M' and P' are the complements in the decompositions defined by M and P respectively. Similarly, the *effect* is defined as

$$\mathrm{eff} \colon M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T^\dagger} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P.$$

Note that both cau and eff send causal states to causal states. Intuitively, these capture the way the previous and next state of P is constrained by the restricted state $s_{|M}$. When the mechanism is the unit $\mathbb{1}$, then the cause and effect $\mathrm{cau}, \mathrm{eff} \colon \mathbb{1} \longrightarrow P$ are respectively called the *unconstrained cause*, denoted $u\mathrm{cau}$ and the *unconstrained effect*, denoted $u\mathrm{eff}$.

Definition 3.16. Given a state s and a mechanism-purview pair M, P, the associated *cause repertoir* is defined to be the state

$$\mathrm{cau}(s, M, P) = \mathbb{1} \xrightarrow{s_{|M}} M \xrightarrow{\mathrm{cau} \otimes u\mathrm{cau}} P \otimes P' \simeq S.$$

One defines an *effect repertoir* $\mathrm{eff}(s, M, P)$ similarly.

We also need to decompose and cut causes and effects.

Definition 3.17. Given a decomposition (M_1, M_2) of a mechanism M, and a decomposition (P_1, P_2) of a purview P, the *decomposed* cause is the process

$$\mathrm{cau}^{P_1,P_2}_{M_1,M_2} \colon M \simeq M_1 \otimes M_2 \xrightarrow{\mathrm{cau}\otimes\mathrm{cau}} P_1 \otimes P_2 \simeq P,$$

and again similarly for defining the *decomposed* effect.

The *decomposed* cause and effect repertoirs are then defined similarly to the normal ones, by replacing cau and eff with their decomposed variants in the definitions. We denote these states by $\mathrm{cau}^{P_1,P_2}_{M_1,M_2}(s, M, P)$ and $\mathrm{eff}^{P_1,P_2}_{M_1,M_2}(s, M, P)$ respectively. Given a cut direction α, the *cut* cause and effect repertoirs $\mathrm{cau}^{\alpha}(s, M, P)$ and $\mathrm{eff}^{\alpha}(s, M, P)$ is similarly defined by using the cut time evolution $T^{\psi,\alpha}$, where ψ is the decomposition associated to the mechanism M. Combining the two approaches yields also *cut-decomposed* cause and effect repertoirs.

For a generalized IIT, one needs to be able to compare causes and effects along different substructures. This is done by assuming that there is a distance function on the states of S, $d(-,-) \colon \mathrm{St}(S) \times \mathrm{St}(S) \longrightarrow \mathbb{R}_{\geqslant 0}$. As in CITE , we do not require this to be a metric, though it usually will be in examples. In classical IIT, this metric is the Wasserstein metric, while in quantum-classical IIT it is the metric induced by the norm of the system C^*-algebra.

We now have enough data to define the IIT-algoritm, as presented in CITE and CITE . The algorithm defines integration for all cause and effect repertoirs, and uses this to aggregate upwards to an integration value for the total system S.

Definition 3.18. Let $\mathrm{cau}(s, M, P)$ be a cause repertoir. Its *integration* $\varphi(\mathrm{cau}(s, M, P))$ is defined as the minimum distance from $\mathrm{cau}(s, M, P)$ to a decomposed repertoir associated to a non-trivial decomposition:

$$\min_{(M_1,M_2),(P_1,P_2)} d(\mathrm{cau}(s, M, P), \mathrm{cau}^{P_1,P_2}_{M_1,M_2}(s, M, P)).$$

We define the integration level of an effect repertoir $\mathrm{eff}(s, M, P)$ similarly.

Definition 3.19. Given a state $s \colon \mathbb{1} \longrightarrow S$ and a subsystem $M \in \mathrm{Sub}_s(S)$, the *core cause* is the sub-system $P^{cc} \in \mathrm{Sub}_s(S)$ such that the associated cause repertoir $\mathrm{cau}(s, M, P)$ has maximal integration. Similarly, the *core effect* is the sub-system P^{ce} that maximizes the integration of the associated effect repertoir $\mathrm{eff}(s, M, P)$.

Definition 3.20. The integration of a mechanism M is then defined as

$$\varphi(M) := \min(\varphi(\mathrm{cau}(s, M, P^{cc})), \varphi(\mathrm{eff}(s, M, P^{ce}))).$$

Given a sub-object $M \in \mathrm{Sub}_s(S)$, the triple consisting of the core cause repertoir, the core effect repertoir and the resulting integration defines the *concept* of M. Allowing M to vary, defines the *Q-shape* of the state s. The collection of all Q-shapes defines the *experience space* $\mathbb{E}(S)$ of the system S. See CITE for the details.

Remark 3.21. One of the fundamental beliefs in IIT is that any conscious experience a system S can have is an element of $\mathbb{E}(S)$.

By minimizing over all possible decompositions, one can similarly define the integration of a Q-shape. The subsystem with maximally integrated Q-shape is the system's *major complex*, which by definition defines an element $\mathbb{E}(S, s)$ in the experience space $\mathbb{E}(S)$. IIT claims that this element is the particular concious experience of the system S in the state s, and the associated Φ-value is defined as the size of this concious experience: $\Phi(S, s) = \|\mathbb{E}(S, s)\|$.

If this Φ-value is zero, then it means that for all mechanisms M, the integration $\varphi(M)$ is zero, meaning that for all purviews P there is a decomposition with the same integration of its cause and effect repertoir. In principle, decomposing the system does not change its cause and effect repertoirs. We now use this to proove our main result.

Lemma 3.22. *Let $(S, \mathbb{D}, T)$ be a system, $s \colon \mathbb{1} \longrightarrow S$ a state of S, (A, B, ψ) a decomposition and α a cut direction. If $\Phi(S, s) = 0$, then for every sub-system $U \in \mathrm{Sub}_s(S)$ the restricted dynamics $T_{|U}$ and the restricted cut-dynamics $T^{\psi,\alpha}_{|U}$ are equal in $\mathcal{F}(U)$.*

Proof. By definition there is a complement U' of U such that (U, U') is a decomposition of S. We consider U as a mechanism, with purview the total system S. The associated cause repertoir $\mathrm{cau}(s, U, S)$ is equivalent to the composite

$$\mathbb{1} \xrightarrow{s_{|U}} U \xrightarrow{T_{|U}} U \xrightarrow{1_U \otimes w_{U'}} U \otimes U' \cong S.$$

The same description holds for the cut-decomposed cause repertoir $cau^{\mathbb{1},S,\alpha}_{U,U'}(s, U, S)$, i.e., it is equivalent to the composite

$$\mathbb{1} \xrightarrow{s_{|U}} U \xrightarrow{T^{\psi,\alpha}_{|U}} U \xrightarrow{1_U \otimes w_{U'}} U \otimes U' \cong S.$$

As the embedding $\mathrm{End}(U) \hookrightarrow \mathrm{End}(U)$ defined by $f \mapsto (\mathrm{Id}_U \otimes w_{U'}) \circ f \circ s_{|U}$ is injective in the image $\mathcal{F}(U)$, equality of the cause repertoirs implies equality of the underlying endomorphisms of U. In particular, the restricted endomorphisms $T_{|U}$ and $T^{\psi,\alpha}_{|U}$ are equal if and only if the associated cause repertoirs are equal.

Now, by assumption we have $\Phi(S, s) = 0$, which by definition means we have non-trivial decompositions of U and S such that either the cause or effect repertoir factorizes. Without loss of generality we assume this to be the cause repertoir: $\mathrm{cau}(s, U, S) = cau^{S_1,S_2}_{U_1,U_2}(s, U, S)$. By construction, the cut time evolutions and the cause and effect maps are defined by inserting noise on complements. Hence, this equivalence on cause repertoirs implies that the action of T on U is the same as the restricted action of $T^{\psi,\alpha}$.

Given a 1-simplex $\sigma\colon U \longrightarrow V$ we then have $\mathcal{F}(\sigma)(T) = \mathcal{F}(T^{\psi,\alpha})$, which by definition implies that $T_{|U}$ and $T^{\psi,\alpha}_{|U}$ are equivalent in $\mathcal{F}(U)$. □

Remark 3.23. Injectivity in the above argument relies crucially on the use of the noise state w rather than the actual system state. In Remark 3.5 we mentioned the option of state-dependent integration, but in this case the above argument fails, and one would need a different approach to make similar claims.

Proof of Theorem 3.13. The assumption of vanishing integrated information $\Phi(S, s) = 0$ implies by definition that for all mechanisms M, we have

$$\varphi(M) = \min(\varphi(\mathrm{cau}(s, M, P^{cc})), \varphi(\mathrm{eff}(s, M, P^{ce}))) = 0.$$

In particular, for all mechanism-purview pairs M, P, there exists associated decompositions $M \simeq M_1 \otimes M_2$ and $P \simeq P_1 \otimes P_2$ such that either

$$\mathrm{cau}(s, M, P) = \mathrm{cau}^{P_1,P_2}_{M_1,M_2}(s, M, P),$$

or

$$\mathrm{eff}(s, M, P) = \mathrm{eff}^{P_1,P_2}_{M_1,M_2}(s, M, P).$$

The goal is to exploit this fact to show that all obstruction classes are coboundaries. To accomplish this, we fix a decomposition $\psi\colon S \simeq A \otimes B$ and a cut direction $\alpha \in \{\rightarrow, \leftarrow, \leftrightarrow\}$. To show that the associated obstruction class $[\Delta^{\psi,\alpha}] = 0$ it suffices to show that it is a coboundary of some 0-cochain θ, in other words that $\Delta^{\psi,\alpha} = \delta\theta$.

We define this cochain by $\theta(U) := T_{|U}$ for all $U \in \mathrm{Sub}_s(S)$. On a 1-simplex $\sigma\colon U \longrightarrow V$ the coboundary is given by

$$\delta\theta(\sigma) = \rho^V_U(T_{|V}) - T_{|U}.$$

As a simplex is an inclusion of a subsystem, Lemma 3.22 gives $\rho^V_U(T_{|V}) = \rho^V_U(T^{\psi,\alpha}_{|V}) = T^{\psi,\alpha}_{|U}$, hence $\delta\theta(\sigma) = \Delta^{\psi,\alpha}(U)$. In particular, $\Delta^{\psi,\alpha} = \delta\theta$, so the obstruction is a coboundary, meaning that we have $[\Delta^{\psi,\alpha}] = 0$ in the integrated cohomology group $H^1_{int}(S)$. □

4 Extensions

Our framework naturally allows for some extensions and future directions, of which we here outline two. As mentioned before, our framework in this paper only uses the binary value of Φ – it is either zero or non-zero – but the true strength of IIT lies in its *size*. Hence our first extension constructs a proper analogue to the Φ-value in our framework, which we denote Φ_{coh}. Further, our cohomological Φ-value, as well as our previously defined obstruction classes, all lie in the first cohomology groups $H^1_{int}(S)$ of the system $(S, \mathbb{D}, T)$. Utilizing the higher order cohomology groups as well, we obtain higher order obstruction classes and higher order cohomological Φ-values, which are analogous to the higher order synergy anf contextuality of CITE .

4.1 Cohomological integrated information

In Section 3.3 we introduced a distance function between states,

$$d\colon \mathrm{St}(S) \times \mathrm{St}(S) \longrightarrow \mathbb{R}_{\geqslant 0},$$

that we used to describe Tull–Kleiner's genealized IIT-algorithm. We here use this to infer a distance function on the cohomology groups, which we then use to define te cohomological integrated information Φ_{coh}.

Definition 4.1. Let $A \in \mathrm{Sub}_s(S)$ be a subsystem. The *distance* between two endomorphisms f, g of A is defined by how differently they act on the local state $s_{|A}$.

In other words,

$$\|f-g\|_A := d(f\circ s_{|A}, g\circ s_{|A}).$$

This extends linearly to a distance function on $\mathcal{F}(A)$. Furthermore, given a k-cochain $c\in C^k(S;\mathcal{F})$, we define its *size* as its largest value on the set of all k-simplicies:

$$\|c\| := \max_{\sigma} \|c(\sigma)\|.$$

In order to infer a size on cohomology classes, we simply need to incorporate the fact that cohomology classes are only defined up to a coboundary, as they are quotient classes of abelian groups. Hence, given a cohomology class $[c]\in H^k_{int}(S)$, we define

$$\|[c]\| := \min_{\theta\in C^{k-1}(S;\mathcal{F})} \|c+\delta\theta\|.$$

This definition makes the value intrinsic, as it minimizes over all possible representations of the class.

Definition 4.2. Let S be a system in a state s. The *cohomological integrated information* is defined as the size of the smalles obstruction class,

$$\Phi_{coh}(S,s) := \min_{\psi,(\alpha_k)} \|[\Delta^{\psi,(\alpha_k)}]\|.$$

The following result immediatly follows from the definition combined with Theorem 3.13.

Proposition 4.3. *If s is a state of a system $(S,\mathbb{D},T)$ such that $\Phi(S,s)=0$, then also $\Phi_{coh}(S,s)=0$.*

We hope to revisit this definition in future work and connect it to the definition of the standard integrated information value $\Phi(S,s)$. For now we explain how this solves Tull and Kleiners search for a functorial construction of integrated information.

In order to have functoriality, we need to understand what categories are present. In particular, we need a category that describes both systems $(S,\mathbb{D},T)$ and their states s. The recipient category will be the poset category $\mathbb{R}_{\geqslant 0}$ of non-negative real numbers. A map into this poset category is functorial when it is monotone.

Definition 4.4. A *system state* in a process theory $\mathcal{C}$, is a tuple $(S,\mathbb{D},T,s)$, where $(S,\mathbb{D},T)$ is a system in $\mathcal{C}$ and $s\colon \mathbb{1}\longrightarrow S$ is a state of S. A *morphism* of system states is a causal map

$$(S,\mathbb{D},T,s)\longrightarrow(S',\mathbb{D}',T',s')$$

that

1. preserves states: $f\circ s = s'$;
2. preserves dynamics: $f\circ T = T'$;
3. preserves decompositions: for a decomposition (A,B,ψ) these exists a decomposition $(f(A),f(B),\psi')$ into subsystems identified by f;
4. induces a functor on the poset categories: $f_*\colon \mathrm{Sub}_s(S)\longrightarrow \mathrm{Sub}_{s'}(S')$;
5. is compatible with the restriction maps: $f(\rho^V_U(c)) = \rho^{f(V)(f(c))}_{f(U)}$;
6. is non-expansive: $\|f_*(c)\|\leqslant\|c\|$.

The resulting category of system states and morphisms is denoted $\mathrm{Sys}(\mathcal{C})$.

It follows from the definition of the category above that the cohomological integrated information is a functor, as the obstruction classes are functorial by design, and the properties of morphisms in $\mathrm{Sys}(\mathcal{C})$ ensures that it is a monotone map. Hence we have the following proposition.

Proposition 4.5. *Let $\mathcal{C}$ be a process theory. The cohomological integrated information is a functor*

$$\Phi_{coh}\colon \mathrm{Sys}(\mathcal{C})\longrightarrow\mathbb{R}_{\geqslant 0}.$$

In particular, isomorphic system states have equal cohomological integrated information,

4.2 Higher order integration

Our setup of using obstruction theory easily allows us to extend integrated information to higher orders – something that cohomology theory is a natural arena for. There are some attempts at this already, se for example CITE , but we will save comparisons for future work.

Higher obstruction classes are defined using more refined decomposition of the system S. The intuition is that, given a decomposition $\psi\colon S\simeq A\otimes B\otimes C$, the Φ-values computed pairwise from these could all be zero, even if the total Φ-value is not. This failure should be measured not by the bipartite obstruction classes previously defined, but by a higher obstruction class.

Definition 4.6. Let S be an object in a process theory $\mathcal{C}$. A *k-ary decomposition* of S is a k-tuple

$(A_1, A_2, \ldots, A_k)$ together with a causal isomorphism

$$\psi \colon S \simeq A_1 \otimes A_2 \otimes \cdots \otimes A_k.$$

We will denote such a decomposition simply by (A_k). In particular, a k-ary decomposition is an iterated binary composition, hence naturally lives in $\mathbb{D}(S)$.

By inserting noise, we can use these to construct *multi-cut* time evolutions $T^{\psi,(\alpha_k)}$, where (α_k) is a sequence of k directions, also allowing for some of the values being empty.

Definition 4.7. Given a system $(S, \mathbb{D}, T)$, a k-ary decomposition (A_k) and a cut-vector (α_k), we define the associated obstruction class on a $k+1$-simplex $\sigma = U_0 \to \cdots \to U_k$ as the cohomology class of

$$\Delta^{\psi,(\alpha_k)}(\sigma) = \rho_{U_0}^{U_k}(T_{|U_k}) - \rho_{U_0}^{U_k}(T_{|U_k}^{\psi,(\alpha_k)}),$$

where $\rho_{U_0}^{U_k}$ is the iterated restriction from U_k to U_0 defined by σ.

This is a class living in $H_{int}^k(S)$, which measures the failure of the largest subsystem U_k to decompose into k parts that are "(α_k)-independent" – if the direction vector consisted of only bidirectional arrows, then these parts would have been independent. For the case $k = 1$, these classes reduce to the ones studied earlier.

Analogously to the construction above for $\Phi_{coh}(S, s)$ we can define higher order cohomological integrated information.

Definition 4.8. Let $(S, \mathbb{D}, T, s)$ be a system state. Its *k-th order cohomological integrated information* is defined as the size of the smallest higher order obstruction class:

$$\Phi_{coh}^k(S, s) := \min_{\psi,(\alpha_k)} \|[\Delta^{\psi,(\alpha_k)}]\|.$$

As in Proposition 4.5, these higher order classes are functorial, and for $k = 1$ we have $\Phi_{coh}^1(S, s) = \Phi_{coh}(S, s)$ as defined above.

This defines for a system state $(S, \mathbb{D}, T, s)$ a sequence of real numbers $\Phi_{coh}^1(S, s), \Phi_{coh}^2(S, s), \Phi_{coh}^3(S, s), \ldots$ that detect higher and higher orders of integration. We round off the paper with the following defintion and conjecture.

Definition 4.9. The integration order of a system state $(S, \mathbb{D}, T, s)$ is defined as the smallest k such that $\Phi_{coh}^k(S, s) \neq 0$. If all higher cohomological integrated information vanishes, i.e., $\Phi_{coh}^k(S, s) = 0$ for all k, then we say it has integration order ∞.

Conjecture 4.10. If $(S, \mathbb{D}, T, s)$ is a system state with integration order ∞, then $\Phi(S, s) = 0$.